BIOC0001 – Proteins and Enzymes

Introduction

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Me



What others say about me?

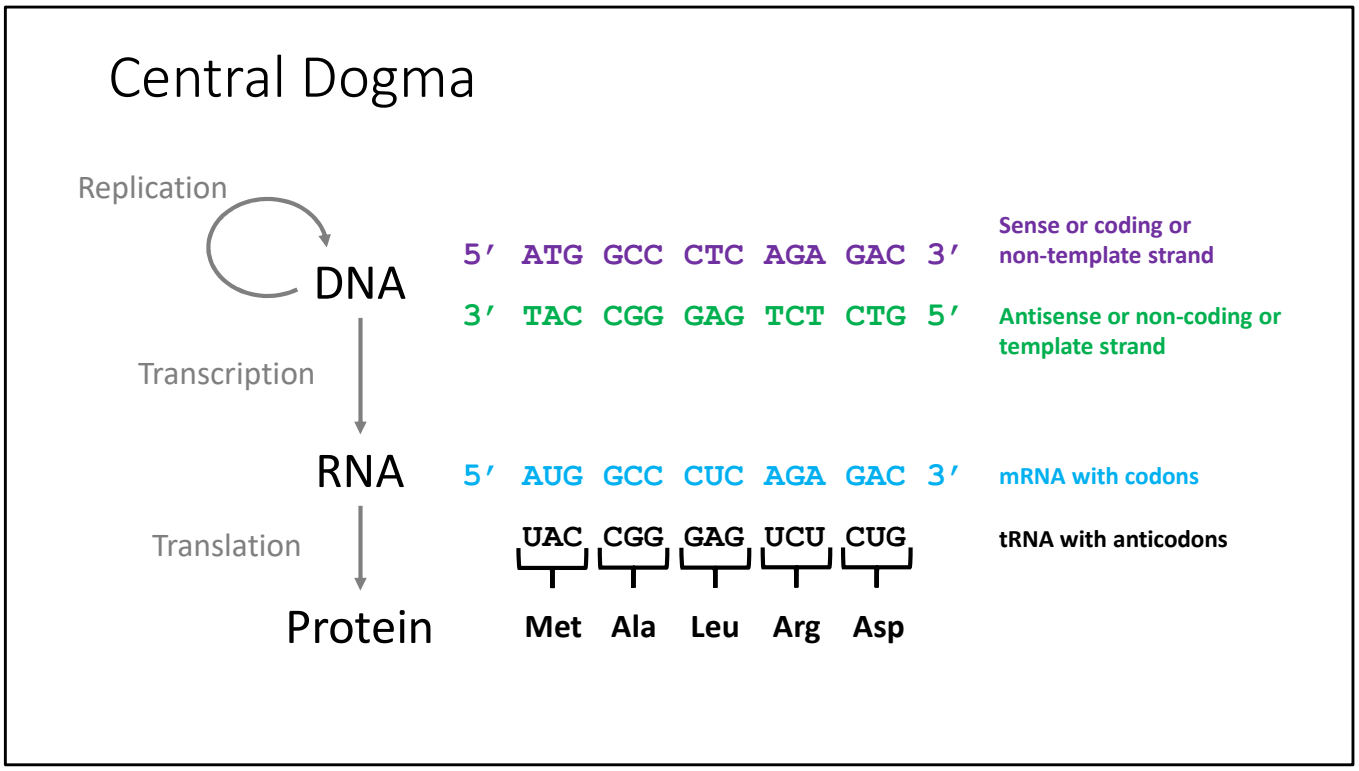
Organised; Hard-working; Supportive; Detail-minded; Creative; Independent; Curious; Introvert ...

What I like?

- Hiking and spending time in the mountains; Running; Going to the theatre; LEGO; Reading comic books; Drawing; Printmaking; Our nephew ...
- Protein structure and function; Enzymes ...
- Looking for “the best way” to teach biochemistry to students ...

My advice?

- Don’t be afraid to change career paths / follow your dreams
- Try out new opportunities so you can find your right fit and you will enjoy working hard
- You can learn anything if you put enough time into it and ask for help
- A lot of the things you will try will fail, figure out why and give it another go. Fear of failure is normal!
- Confidence comes with time and experience
- ...



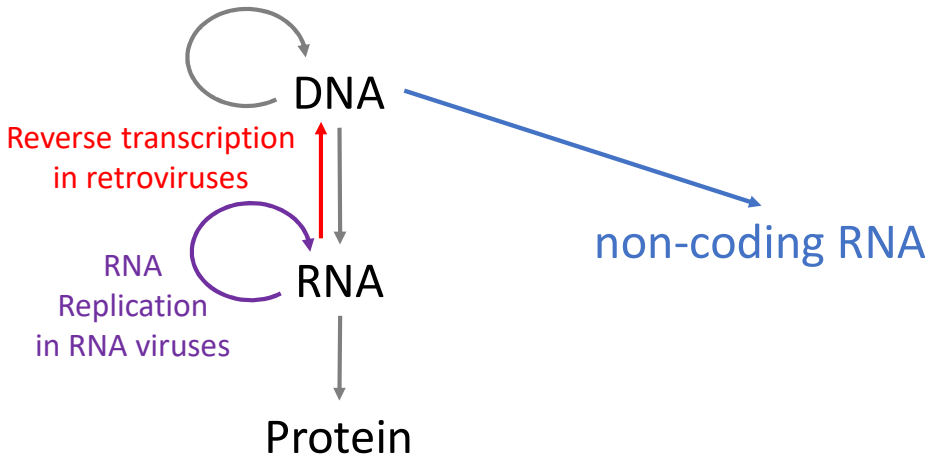
The **central dogma** describes the two-step process, **transcription** and **translation**, by which **genetic information** is converted into **proteins**.

- In transcription, the **DNA** sequence of a gene is copied to make an **RNA** molecule. In eukaryotes, the RNA molecule must undergo processing to become a mature **messenger RNA** (**mRNA**).
- In translation, the sequence of the mRNA is decoded with the help of **transfer RNA (tRNA)** to specify the **amino acid sequence** of a protein, the **primary structure**.

DNA is double-stranded, but only one strand serves as a **template** for transcription at any given time. This template strand is called the **non-coding or antisense** strand. The **non-template** strand is referred to as the **coding or sense** strand because its sequence will be the same as that of the new RNA molecule.

The mRNA encodes a protein as a series of **codons**. Each codon is recognized by a specific tRNA molecule with the corresponding **anticodon**. Each codon corresponds to a specific amino acid or **stop signal**.

Alternative Pathways for the Central Dogma



The flow of information as stated in the central dogma has multiple pathways!

The general transfers include DNA → DNA (DNA-dependent DNA polymerase), DNA → RNA (DNA-dependent RNA polymerase), and RNA → protein (ribosome and tRNA).

Special transfers include

- **reverse transcription**: RNA → DNA (RNA-dependent DNA polymerase or reverse transcriptase, in retroviruses) followed by DNA → RNA, and RNA → protein
- **RNA replication**: RNA → RNA (RNA-dependent RNA polymerase, in RNA viruses) followed by RNA → protein

There are also **non-coding RNAs (ncRNA)**. These are transcribed from DNA but not translated into proteins. They play crucial roles in *e.g.* regulating gene expression.

Codons

1st position	2nd position U	2nd position C	2nd position A	2nd position G	3rd position
U	Phe	Ser	Tyr	Cys	U
	Phe	Ser	Tyr	Cys	C
	Leu	Ser	stop	stop	A
	Leu	Ser	stop	Trp	G
C	Leu	Pro	His	Arg	U
	Leu	Pro	His	Arg	C
	Leu	Pro	Gln	Arg	A
	Leu	Pro	Gln	Arg	G
A	Ile	Thr	Asn	Ser	U
	Ile	Thr	Asn	Ser	C
	Ile	Thr	Lys	Arg	A
	Met	Thr	Lys	Arg	G
G	Val	Ala	Asp	Gly	U
	Val	Ala	Asp	Gly	C
	Val	Ala	Glu	Gly	A
	Val	Ala	Glu	Gly	G
	Amino Acids				

Start codon

64 codons
20 amino acids

Codon degeneracy

Figure: Genetic Code (2020)

There are 64 possible permutations, or combinations, of three-letter nucleotide sequences that can be made from the four nucleotides. Of these **64 codons**, 61 represent amino acids, and three are stop signals.

Although each codon is specific for only one amino acid (or one stop signal), the genetic code is described as **degenerate**, or **redundant**, because a single amino acid may be coded for by more than one codon. **Twenty different amino acids** are commonly found in proteins.

Methionine is specified by the **codon AUG**, which is also known as the **start codon**. Consequently, methionine is the first amino acid to dock in the **ribosome** during the synthesis of proteins.

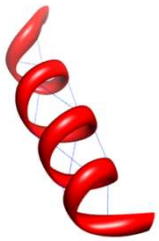
Levels of Organization

Primary → **Secondary** → **Tertiary** → **Quaternary**

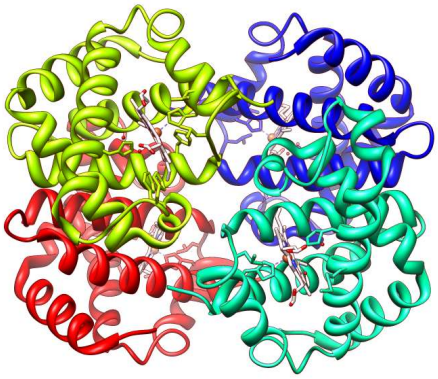
Primary	Secondary	Tertiary	Quaternary
the sequence of amino acids in the polypeptide chain (N to C)	helices, sheets, turns = local, hydrogen-bonded configurations of the polypeptide chain	spatial arrangement of secondary structures in 3D	non-covalent association of tertiary structures (*subunits*) into a complex



...GTLTKV...
...Gly-Thr-Leu-Thr-Lys-Val...



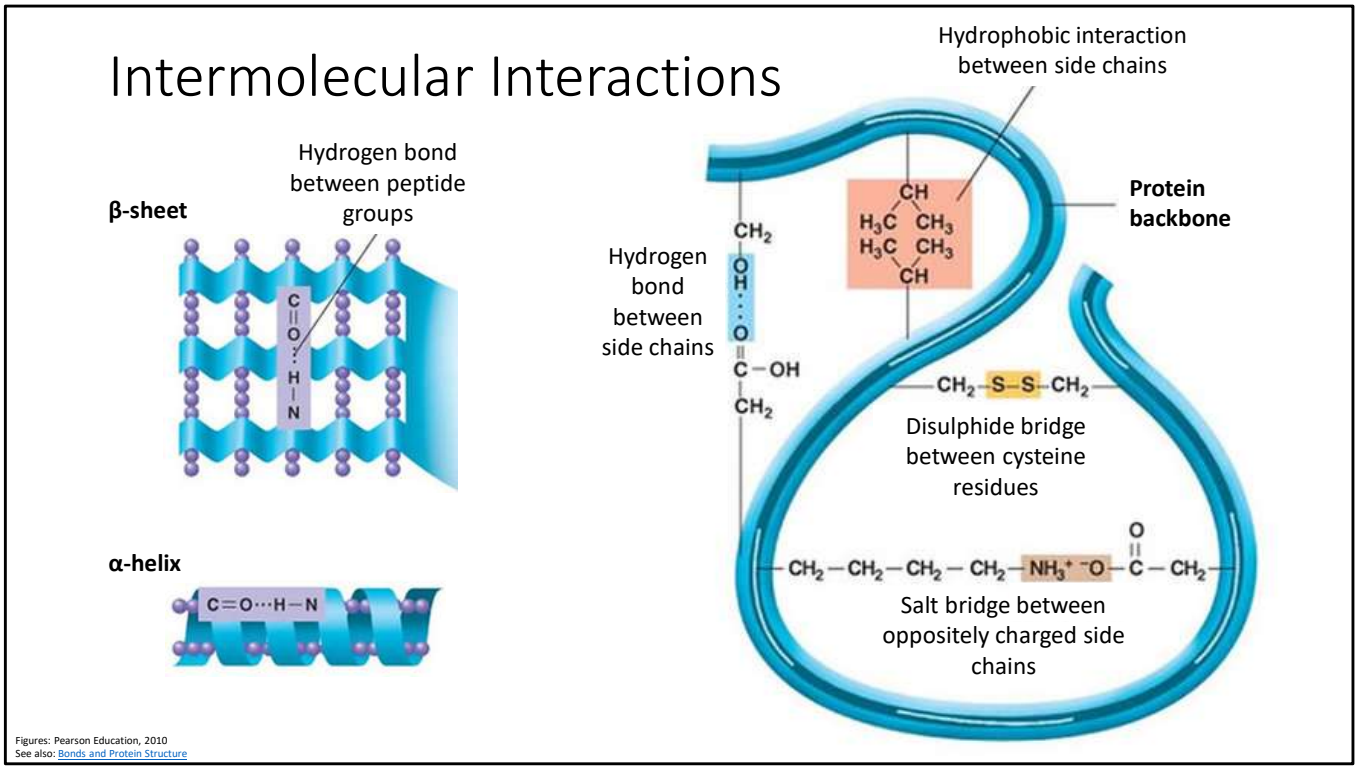




Figures: Protein Data Bank: what-is-a-protein-pres.pptx (live.com) and Intro-to-Biol-Mol-0.pptx (live.com)

- The **primary structure** of a protein is its linear sequence of **amino acid**s. The amino acids are connected by covalent peptide bonds. The **side chain (R-group)** gives each amino acid distinctive structural and chemical properties (size, shape, polarity, charge and hydrophobicity).
- A protein's **secondary structure** (*i.e.* **α-helices, β-sheets, and β-turns**) is the local spatial arrangement of its **backbone** atoms without taking the conformations of its side chains into consideration. It relies on interactions between backbone atoms of predominantly neighbouring amino acids in the sequence.
- A protein's **tertiary structure** refers to its 3D structure. It includes interactions between backbone and sidechain atoms of amino acids further apart in the sequence. It is determined by its primary structure.
- Many proteins are composed of two or more polypeptide chains, referred to as **subunits**. A protein's **quaternary structure** refers to the spatial arrangement of its subunits. Each of the subunits has its own primary, secondary, and tertiary structure.

Shown are the levels of protein structure organization applied to **myoglobin and hemoglobin**. Myoglobin is formed from eight α-helices surrounding an iron-containing **heme**, which binds oxygen. Hemoglobin has four subunits: two identical **α chains** (identical in primary, secondary and tertiary structure) and two identical **β chains** (but different from the α chains); it is a **hetero-tetramer**. All of hemoglobin's subunits are very similar to myoglobin.



The **side chains (R-groups)** and the **backbone** of a protein chain can interact in different ways.

Non-covalent interactions include:

- **hydrogen bonds** between side chains, between peptide groups, between a side chain and a peptide group, and with water;
- **salt bridges (also called charge-charge or ionic interactions)** between oppositely charged amino acids;
- **hydrophobic interactions** between hydrophobic, nonpolar side chains;
- **van der Waals interactions** - weak attractions between atoms in close contact.

Individual non-covalent interactions are generally weak, but together they help determine the overall **shape and stability** of a protein.

Disulphide bridges are **covalent** bonds that form between the sulphur atoms of two cysteine residues. They are very common in extracellular proteins.

Have a look at the *"An Introduction to Chemical Bonds and Protein Structure"* from the *"Online Macromolecular Museum"* on https://earth.callutheran.edu/Academic_Programs/Departments/BioDev/omm/jsmolnew/bonding/chymo.html. It shows examples of the types of forces that play important roles in determining and stabilizing 3D protein structure. This material is only for your interest and to help your understanding of forces in protein structure - there won't be an exam question on chemical bonds in bovine gamma-chymotrypsin!

Sequence-Structure-Function Relationship

Disorder

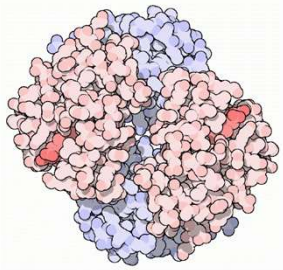
IDPs (see later)

Sequence → **Structure** → **Function**

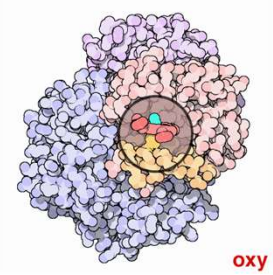
```
DNA      ... CCT GAG GAG ...
             ||| ||| |||
         ... GGA CTC CTC ...

RNA      ... CCU GAG GAG ...

Protein ... Pro Glu Glu ...
```



What it looks like and how it is organized



What it does and how

Figures and video: PDB-101: Molecule of the Month: Hemoglobin

The **sequence-structure-function relationship** states that a protein's amino-acid sequence determines its 3D structure (or ensemble of structures), which in turn determines its biological function(s).

- The specific sequence of amino acids in a polypeptide chain is crucial. Different amino acids have distinct chemical properties, which contribute to the overall structure and function of the polypeptide.
- Any disruption to this structure, through mutations (see next slide) or denaturation (see later), can lead to a loss of function.

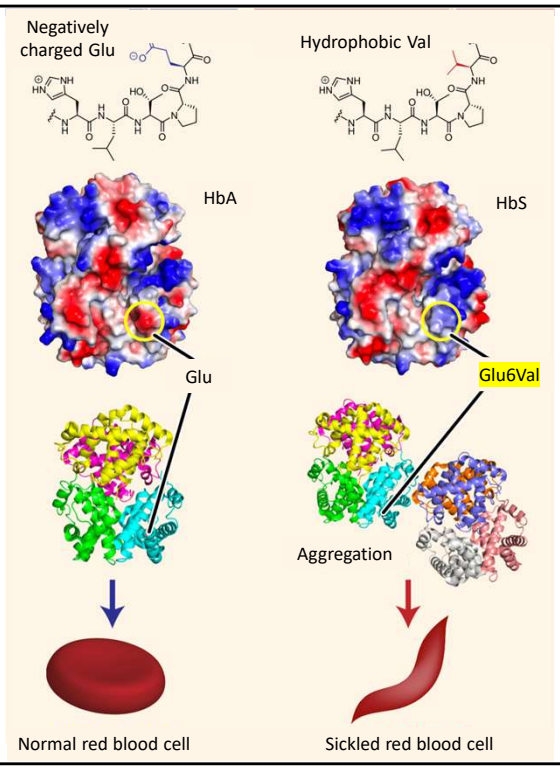
Shown is hemoglobin, made of four subunits, each with a heme group containing an iron atom that binds oxygen. Oxygen binding at the four heme sites does not happen simultaneously. Once one subunit binds oxygen, it introduces small changes in the structure of the corresponding subunits. These changes make it easier for them to bind oxygen. We call this **cooperative binding**. This provides a great advantage. When blood is in the lungs, where oxygen is plentiful, oxygen easily binds to the first subunit and then quickly fills up the remaining ones. Then, as blood circulates through the body, the oxygen level drops while that of carbon dioxide increases. Now, hemoglobin releases its bound oxygen. As soon as the first oxygen molecule is released, the protein starts changing shape, releasing the remaining three oxygens quickly.

Mutations

DNA ... CCT GTG GAG ...
... GGA CAC CTC ...

RNA ... CCU GUG GAG ...

Protein ... Pro Val Glu ...

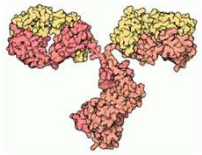


Figures : LabXchange

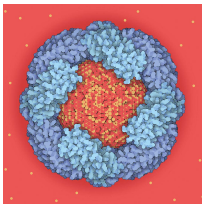
The replacement of an amino acid in the sequence, a **mutation**, can have structural consequences on the resulting protein and thus has a potential effect on its function.

Shown is **sickle cell hemoglobin (HbS)** resulting from a point mutation in the β chain at position 6, in which the negatively charged glutamic acid is replaced by a hydrophobic valine. This change allows the deoxygenated form of the hemoglobin to stick to each other, producing stiff fibers of hemoglobin inside red blood cells. This in turn deforms the red blood cell, which is normally a smooth disk shape, into a C or sickle shape. The distorted cells are fragile and often rupture, leading to loss of hemoglobin. They can also block capillaries.

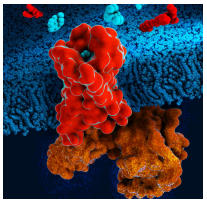
Structures and Functions are Diverse



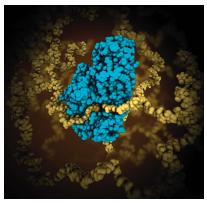
DEFENCE
e.g. antibodies



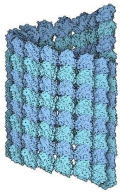
STORAGE
e.g. ferritin storing ion atoms



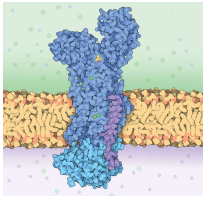
COMMUNICATION
e.g. enkephalins binding to opioid receptors to mediate pain signals



ENZYMES
e.g. α-amylase (saliva) breaking down carbohydrates



STRUCTURAL
e.g. microtubules, tracks for dynein and kinesin



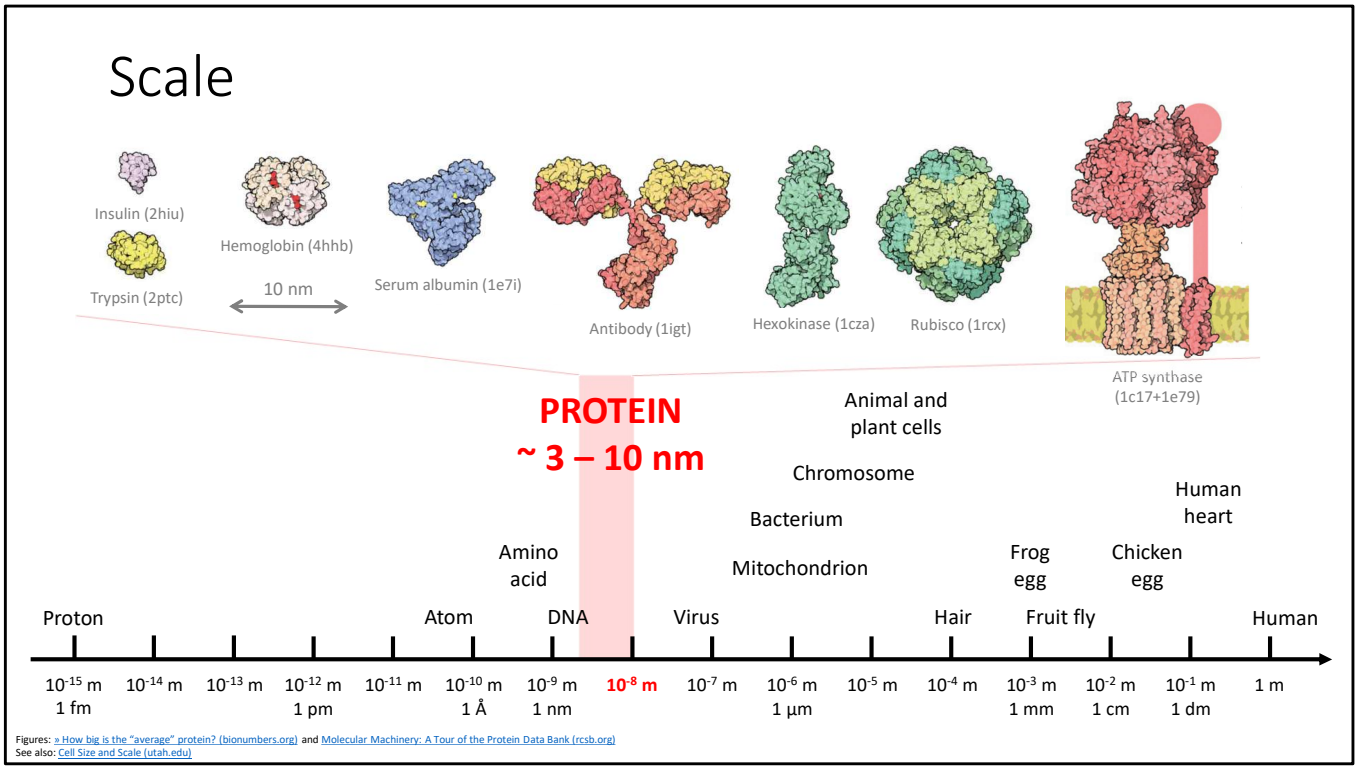
TRANSPORT
e.g. sodium-potassium pumps using ATP to pump ions across a membrane

Figures: Protein Data Bank: 2019-calendar-web.pptx (live.com), 2020-calendar.pptx (live.com), https://pdb101.rcsb.org/motm/175, and https://pdb101.rcsb.org/motm/21

Proteins perform a wide variety of essential functions in life, and their structures are finely tuned to enable these roles:

- **Defence:** Antibodies protect the body from infection. They are Y-shaped molecules with two antigen binding sites, one at the tip of each arm of the Y. They participate in various activities (*e.g.* bind to phagocytic cells or activate the complement system) that are mediated by the tail of the Y.
- **Storage:** Some proteins act as storage molecules. *E.g.* ferritin stores iron atoms safely until the body needs them. This prevents free iron from causing oxidative damage.
- **Communication:** Proteins are also involved in signalling between cells. *E.g.* enkephalins bind to opioid receptors in the nervous system to help mediate pain signals.
- **Enzymes:** Enzymes are biological catalysts that speed up chemical reactions. *E.g.* α-amylase in saliva breaks down carbohydrates into smaller sugars.
- **Structural:** Proteins can provide shape, support, and tracks for movement. *E.g.* microtubules form part of the cytoskeleton and serve as "tracks" for motor proteins like dynein and kinesin to move cargo within cells.
- **Transport:** Some proteins move molecules across membranes. *E.g.* sodium-potassium pumps use energy (ATP) to move sodium and potassium ions in and out of cells, maintaining essential gradients.

Many proteins can fall into more than one of the groups listed. For example, many receptors (communication) are also kinases (enzymes).

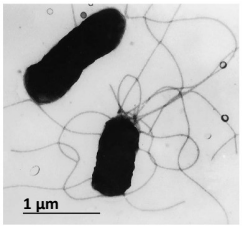


Biological processes occur across a vast range of sizes: from protons (10^{-15} m) to humans (~1 m). **Proteins** typically range from **3 to 10 nanometers (nm)** in size.

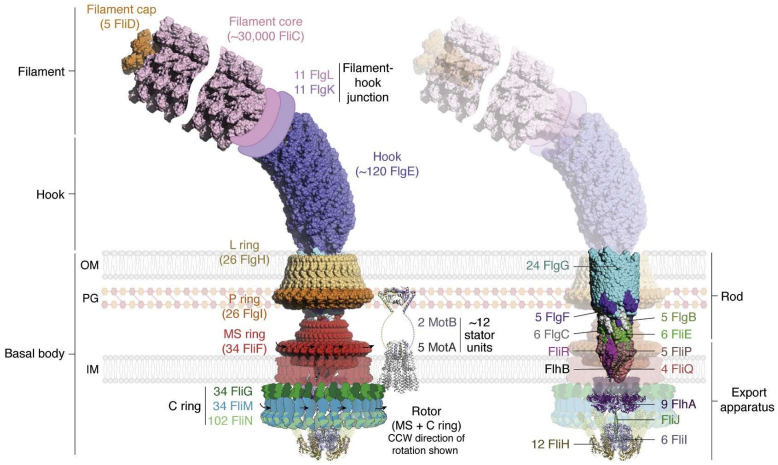
Have a look at the interactive *"Cell size and Scale"* from the *"University of Utah"* on https://learn.genetics.utah.edu/content/cells/scale/.

Higher-Order Complexes

PROTEINS ASSEMBLE INTO HIGHER-ORDER COMPLEXES CRUCIAL TO LIFE



Flagella of *Escherichia coli* observed in transmission electron microscope. They enable the cells to swim at high speeds (~ 100 µm/s) in liquid environments. Eisenbach (2011).



Schematic of a bacterial flagellum comprised of more than 25 different types of proteins, some of them present more than thousands times. Mondino et al (2022).

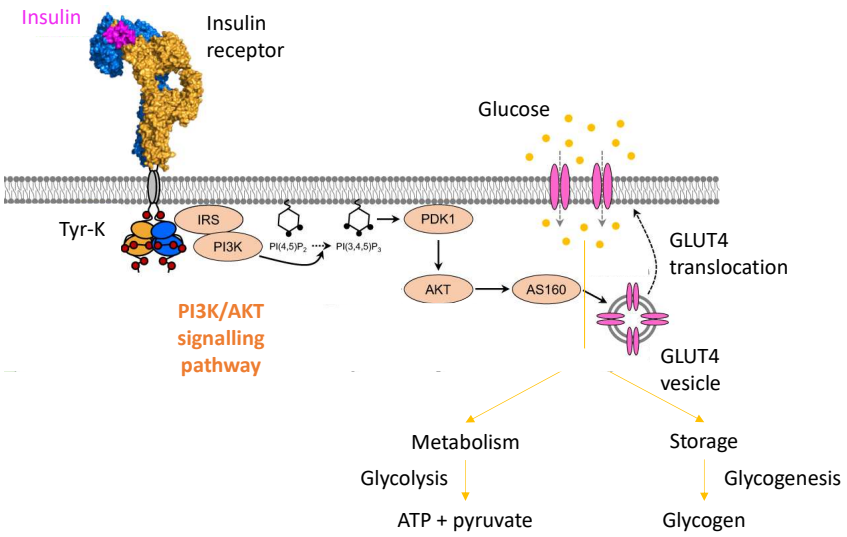
Figures: Bacterial Chemotaxis - Eisenbach - Major Reference Works - Wiley Online Library and 3D cryo-EM imaging of bacterial flagella: Novel structural and mechanistic insights into cell motility - Journal of Biological Chemistry (jbc.org)

Proteins often **assemble into large complexes** to perform specialized tasks. These assemblies are highly organized, with specific protein components arranged in precise structures.

The **bacterial flagellum** is a prime example of a higher-order complex.

Protein Encounters

PROTEINS COME TOGETHER TO CARRY OUT PROCESSES CRUCIAL TO LIFE



Figures: PDB-101: Diabetes Mellitus: Diabetes Mellitus: undefined: Insulin Receptor and A stepwise activation model for the insulin receptor | Experimental & Molecular Medicine

Proteins also come together. These **interactions** often begin at the cell membrane, where external signals (like hormones) trigger internal responses.

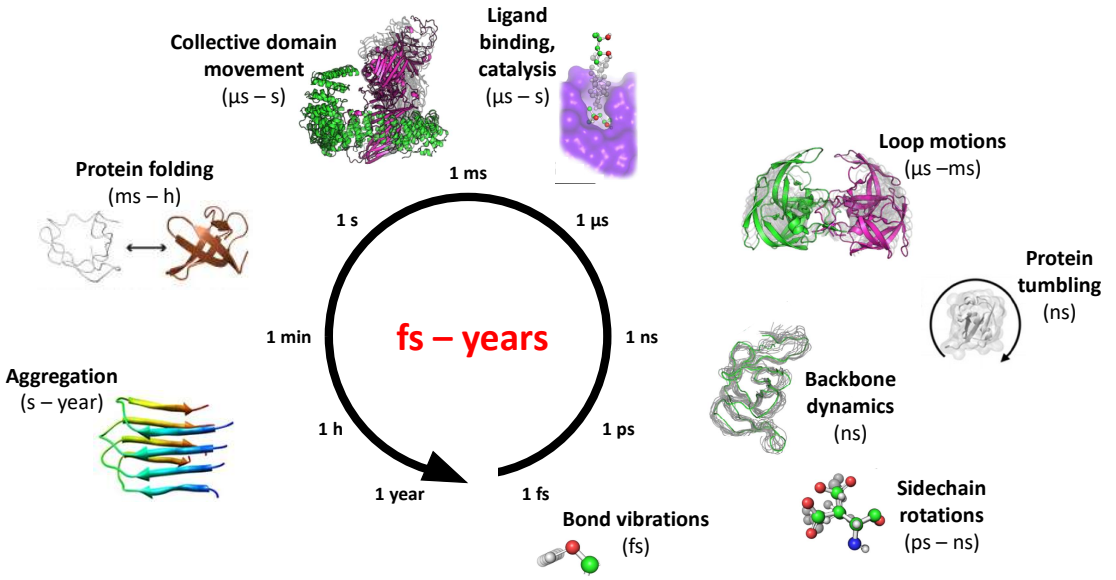
Insulin signalling is a lovely example. The **insulin receptor** is a large, multi-domain, transmembrane protein with both extracellular and intracellular domains. Domains positioned outside the cell bind insulin. Conformational changes resulting from insulin binding activate the tyrosine kinase (Tyr-K) catalytic domain located within the cell through autophosphorylation. The activated Tyr-K in turn phosphorylates other proteins, such as insulin receptor substrate (IRS), which activate proteins downstream in the signalling cascade: phosphoinositide 3-kinase (PI3K), phosphoinositide-dependent kinase-1 (PDK-1), protein kinase B / AKT, and AKT substrate 160 kDa (AS160). This process enables translocation of glucose transporters (GLUT4) to the cell surface, where they contribute to glucose uptake. Following entry into cells, the glucose is metabolized for energy or converted to glycogen for storage.

Problems with insulin signalling can impair the proper management of glucose levels in the blood, leading to the widespread disease **diabetes**. There are two common ways that this happens.

- Type I diabetes is caused by problems with insulin: in some cases, the pancreatic cells that produce insulin are destroyed by autoimmunity, and in other cases insulin is mutated and inactive. This often occurs early in life and requires treatment with insulin to replace the missing insulin.
- Type II diabetes, on the other hand, most often occurs later in life, and is caused by an

acquired resistance to the action of insulin on its receptor. The details are complex and involve phosphorylation of the receptor and its substrates, modifying their action in insulin signalling. The condition is treated with careful attention to diet, lifestyle and medication.

Protein Dynamics



Figures: Protein structure and dynamics - Repository of the Academy's Library (mtak.hu) ; Advanced NMR spectroscopy methods to study protein structure and dynamics – ScienceDirect ; IJMS | Free Full-Text | Dynamics, a Powerful Component of Current and Future in Silico Approaches for Protein Design and Engineering (mdpi.com) ; Probing Biological Water Using Terahertz Absorption Spectroscopy | IntechOpen

Because most of the forces that stabilise tertiary structure are non-covalent, these weak interactions can break and reform readily. **Protein dynamics** can be versatile and covers a large range of movement amplitudes and timescales.

Sub-angstrom vibrations of covalent bonds are the fastest of these movements (fs). Then come side-chain rotations (ps–ns) and backbone fluctuations (ns), followed by loop motions (ns–ms), ligand binding/unbinding events (>100 ns), and collective domain movement (>µs).

IDPs

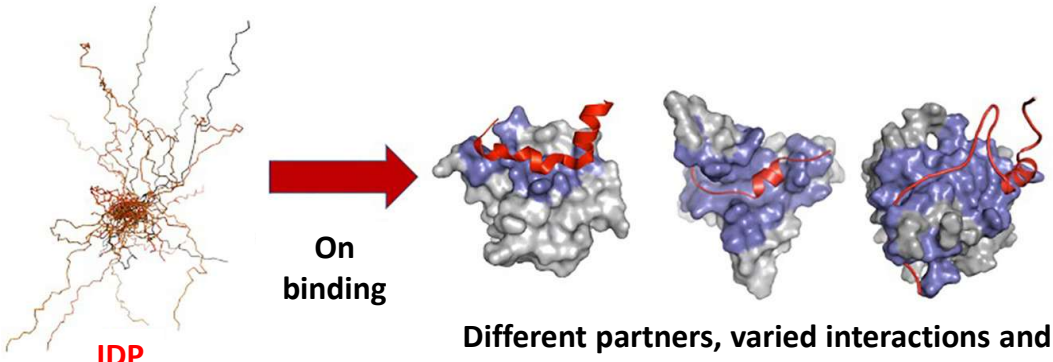
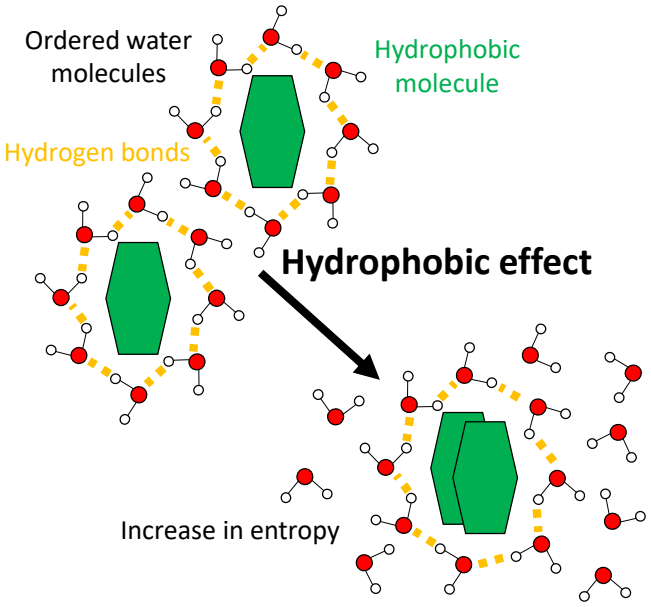


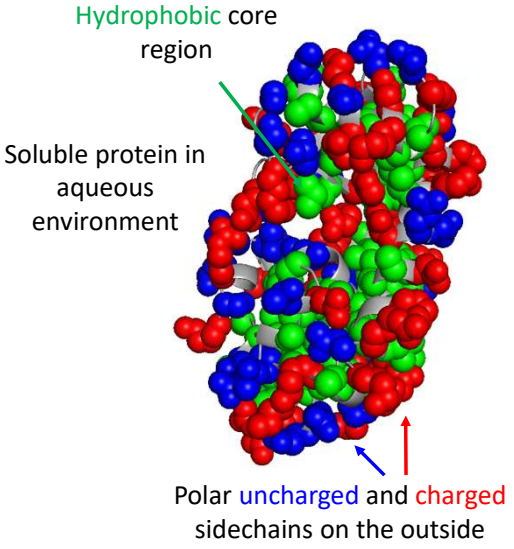
Figure: Intrinsically disordered proteins/regions and insight into their biomolecular interactions - ScienceDirect

An example of extreme protein dynamics is **Intrinsically Disordered Proteins (IDPs)**. In the traditional view of proteins, a polypeptide's function is determined by its specific and relatively stable 3D structure. In recent years, however, it has become apparent that many proteins are in fact partially or completely unstructured. These proteins are called IDPs.

- More than **30% of eukaryotic proteins** are IDPs! They regulate key biological processes by binding to multiple protein partners.
- IDPs are **flexible and malleable**, adapting to structurally different partners, making them ideal "**hub proteins**" in cellular networks!
- IDPs are also thought to be involved many **human diseases**, including cancer (*e.g.* c-Myc and p53), Alzheimer's (*e.g.* tau and amyloid-β), and Parkinson's disease (*e.g.* α-synuclein).
- IDPs usually have amino acid sequences that are **rich in polar and charged residues** and **low in hydrophobic residues**. This prevents them from folding into a stable 3D structure like most other proteins.

Protein Folding and the Hydrophobic Effect





PDB: 1LYD, T4 Lysozyme

Proteins typically fold in aqueous environments, and the **hydrophobic effect** plays a central role in guiding this process. Hydrophobic, nonpolar molecules tend to avoid contact with water. When dispersed in water, they cause surrounding water molecules to form ordered cages, which is entropically unfavourable.

To minimize this ordering, hydrophobic groups aggregate, releasing water molecules and **increasing entropy**. This entropic gain is a key driving force behind protein folding.

As a protein adopts its tertiary structure, **nonpolar side chains are buried in the interior**, shielded from water, while **polar and charged side chains are exposed on the surface**, interacting with the aqueous environment.

This behaviour underpins the distinction between **soluble** and **membrane proteins**. In soluble proteins, hydrophilic residues dominate the surface, while hydrophobic residues form the core. This arrangement ensures solubility and structural stability in water.

Thermodynamics of Protein Folding

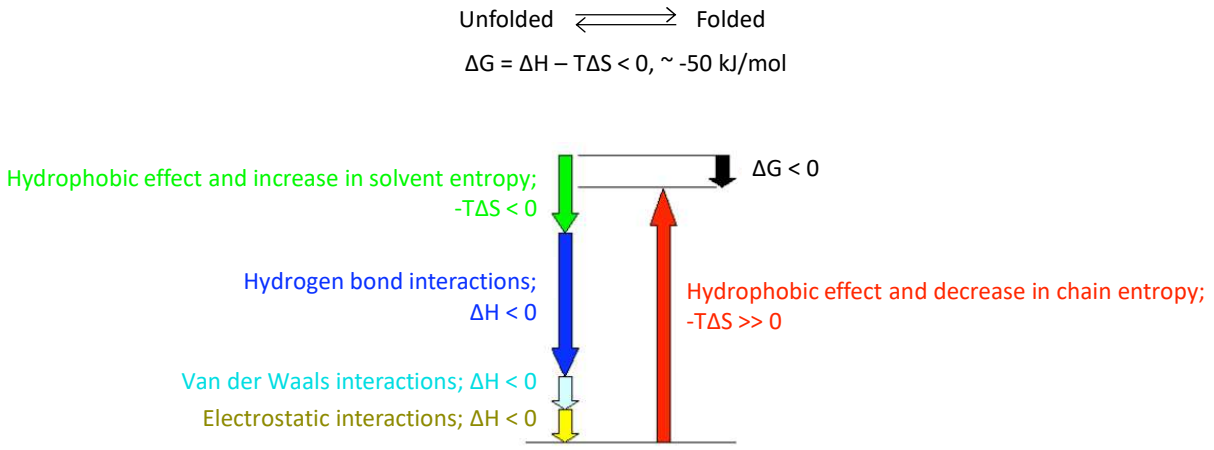


Figure: James Chou, Harvard Medical School

Protein folding is governed by **thermodynamics**, specifically the balance between **entropy (S)** and **enthalpy (H)**, which together determine the **free energy (G)** of the system: $\Delta G = \Delta H - T\Delta S$, where T is temperature and Δ refers to final-initial, *e.g.* $\Delta S = S_{folded} - S_{unfolded}$.

- Entropy is a measure of **disorder** of the system, which is associated with conformational freedom. *E.g.* when a protein folds, its backbone and sidechains lose freedom; this is an entropy cost.
- Enthalpy or heat energy is the **internal energy** of the system from *e.g.* electrostatic interactions, hydrogen bond interactions, van der Waals interactions, hydrophobic interactions, disulphide bridges…

The free energy difference between the native, folded state and the denatured, unfolded state is determined by both entropy and enthalpy: Protein folding involves an increase in solvent entropy due to the burial of hydrophobic moieties and a negative enthalpy contribution due to the formation of favourable interactions. Both compensate for the protein entropy decrease as sidechains and backbones are restricted in the folded state.

Membrane Proteins

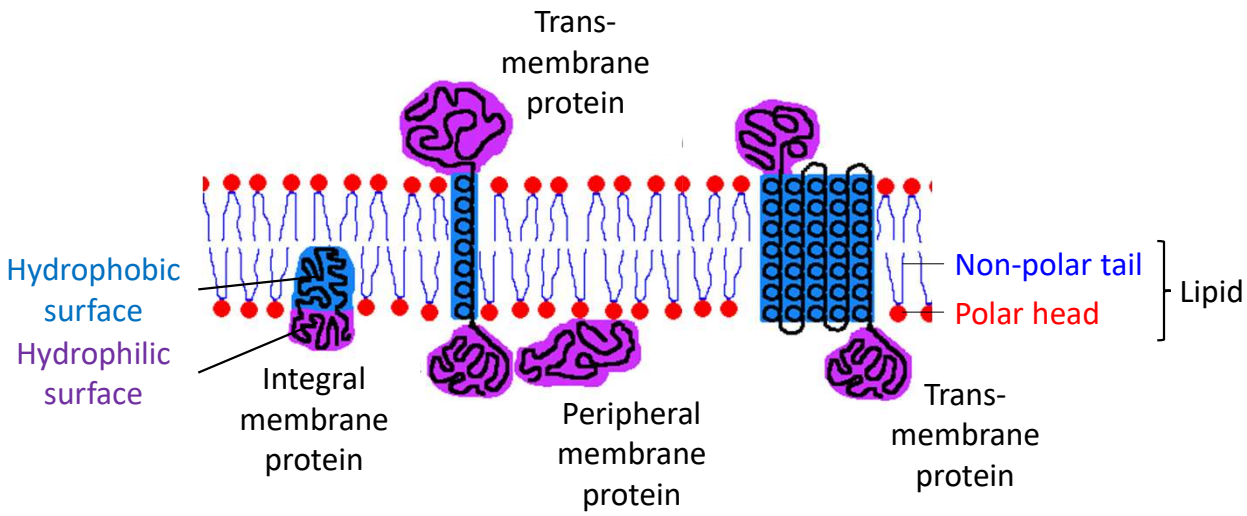


Figure: http://www.cryst.bbk.ac.uk/PPS95/course/7_tertiary/membrane.html

The **phospholipid bilayer** of cell membranes presents a unique environment: a **hydrophobic interior** formed by nonpolar lipid tails, and **hydrophilic surfaces** formed by polar head groups.

Membrane proteins are adapted to this environment through their hydrophobic and hydrophilic regions, which determine their localization and orientation:

- **Integral membrane proteins** have hydrophobic domains that embed within the bilayer's core. These regions interact favourably with the lipid tails, while hydrophilic domains remain exposed to the aqueous surroundings.
- **Transmembrane proteins** span the entire bilayer. Their structure reflects the bilayer's polarity: hydrophobic segments traverse the membrane, and hydrophilic domains extend into the cytosol and extracellular space.
- **Peripheral membrane proteins** do not penetrate the hydrophobic core. Instead, they associate with the membrane surface via **ionic interactions** with polar head groups or other membrane proteins. Their surfaces are predominantly hydrophilic.

Protein Folding in the cell

Chaperones and the proteasome

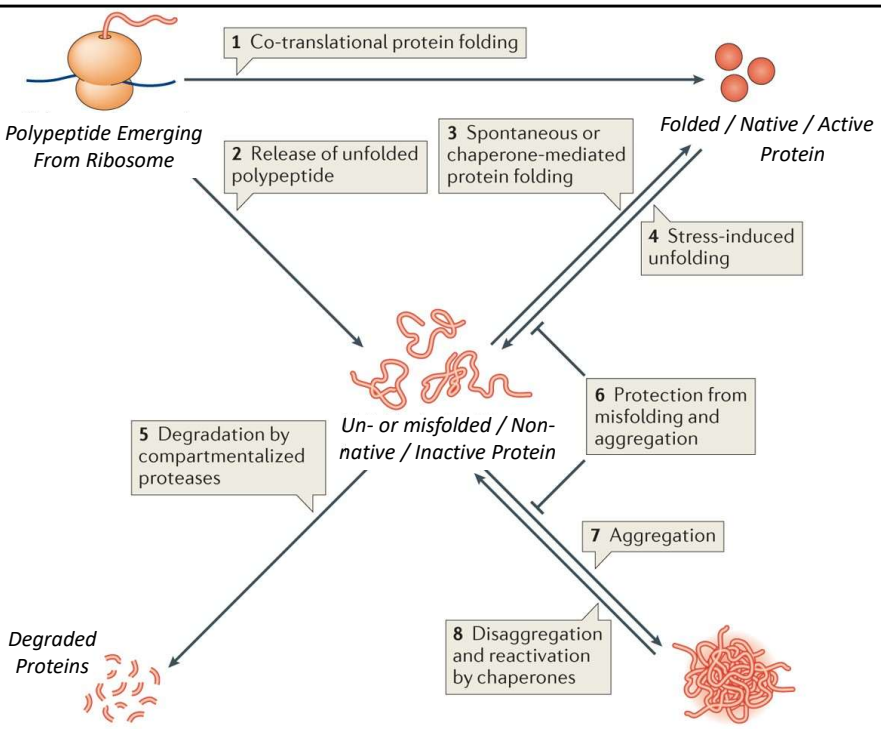


Figure: Protein rescue from aggregates by powerful molecular chaperone machines | Nature Reviews Molecular Cell Biology
See also: PDB-101: Molecule of the Month: Chaperones, PDB-101: Molecule of the Month: Hsp90 and PDB-101: Molecule of the Month: Proteasome

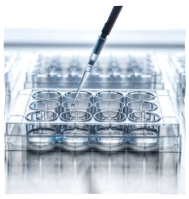
In a cell, most proteins need to **fold** into a correct 3D conformation during and following synthesis on the **ribosome**. Most protein chains fold spontaneously into their final native / active / folded structure, driven by the **hydrophobic effect**. But some – mostly large proteins or proteins with several domains – do not fold spontaneously.

Chaperones are a group of proteins that interact with non-native proteins, *i.e.* unfolded / partially folded or improperly folded proteins. They assist in protein folding by preventing non-specific aggregation between proteins by binding to hydrophobic regions to prevent aggregation or by providing a microenvironment to promote folding. There are several families of chaperones and each possesses different functions. Examples include the **"heat shock proteins" (Hsps)** – see https://pdb101.rcsb.org/motm/32 for more information on Hsp60 and Hsp70 and see https://pdb101.rcsb.org/motm/108 for more information on Hsp90.

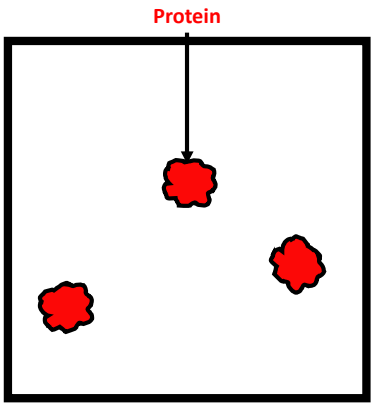
Protein folding is a vital process. **Misfolded proteins** often have hydrophobic amino acids on their surfaces, instead of tucked safely inside. These hydrophobic patches can associate strongly with similar patches on other proteins, forming large **aggregates**. Protein aggregation can lead to protein conformational diseases such as **Alzheimer's, Parkinson's, Huntington's, sickle cell anaemia, and mad cow disease**. Apart from chaperones assisting proteins to fold, another line of cell defence against misfolded proteins is called the **proteasome**. Misfolded proteins will be targeted for destruction by the proteasome, the cell's recycling centre. See https://pdb101.rcsb.org/motm/166 for more information.

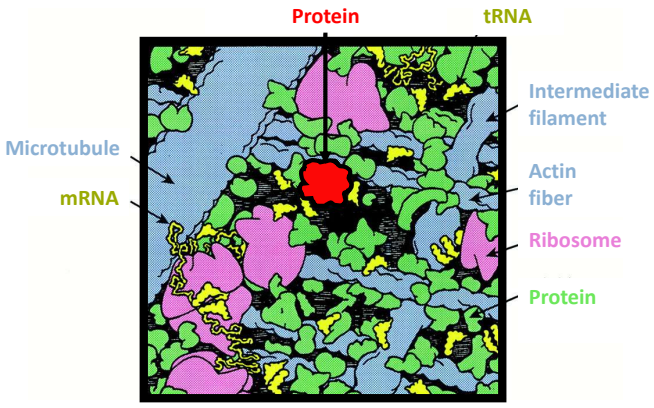
In Silico, In Vitro (Acellular / Cellular), In Vivo!











Figures: MS Office Stock Images and The Influence of Macromolecular Crowding and Macromolecular Confinement on Biochemical Reactions in Physiological Media – ScienceDirect

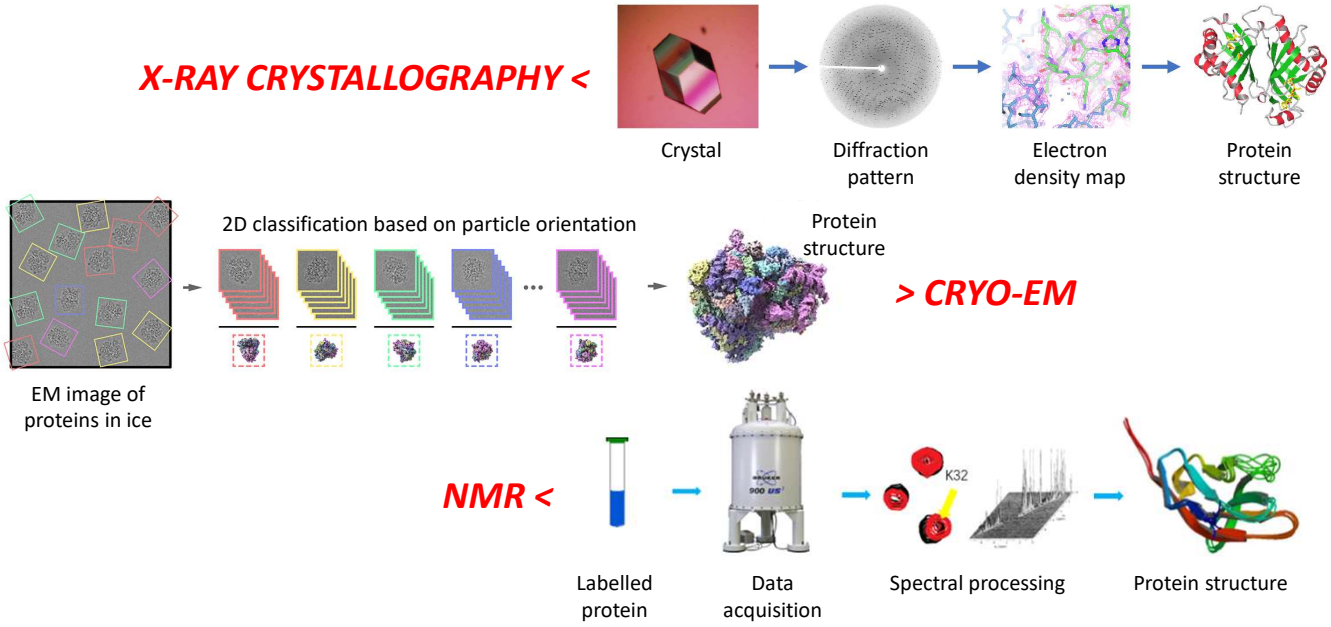
We study proteins in **different experimental** environments, from isolated systems to the complexity of living cells:

- ***In Silico***: Computer simulations and modelling
- ***In Vitro***: Experiments outside living organisms, *e.g.* purified proteins, cell cultures
- ***In Vivo***: Studies within living organisms.

Studying proteins in isolation is useful because it allows precise measurement of folding, binding, and activity, but **real biological function** emerges in context.

Understanding the **limitations and strengths** of each approach helps us interpret experimental data more accurately.

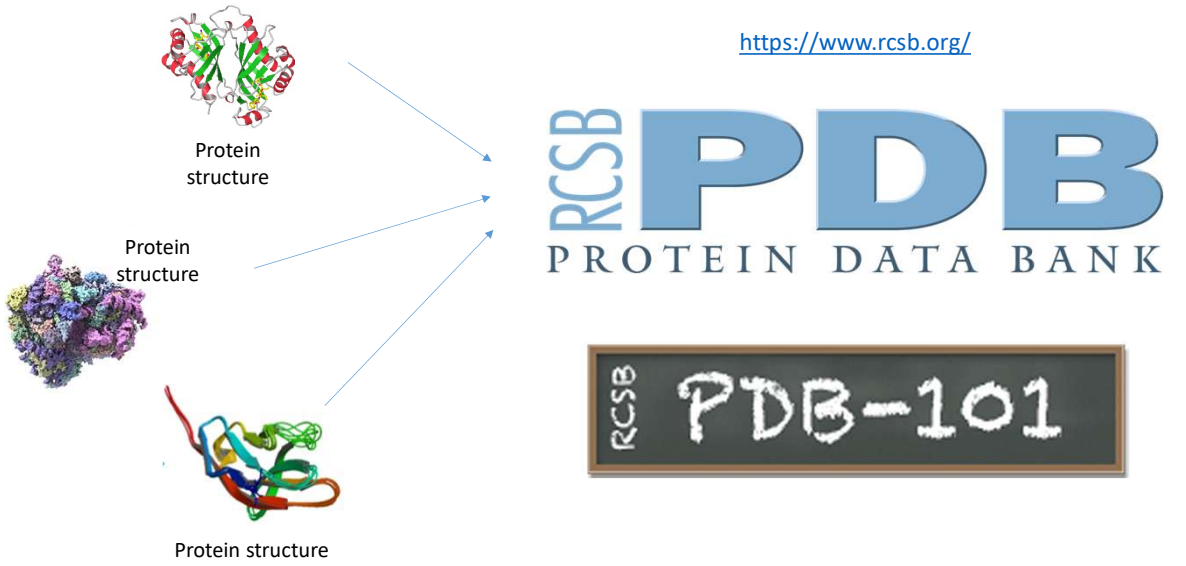
After Cloning, Expression, and Purification...



Figures: X-ray Crystallography Platform - Creative Biostructure, Cryo-EM Explained: Visualizing Life's Machinery - Morgridge Institute for Research, and Nuclear Magnetic Resonance (NMR) Platform - Creative Biostructure

The known protein structures have come to light through **X-ray crystallographic**, **nuclear magnetic resonance** (**NMR**) or **cryo-electron microscopy (cryo-EM)** studies.

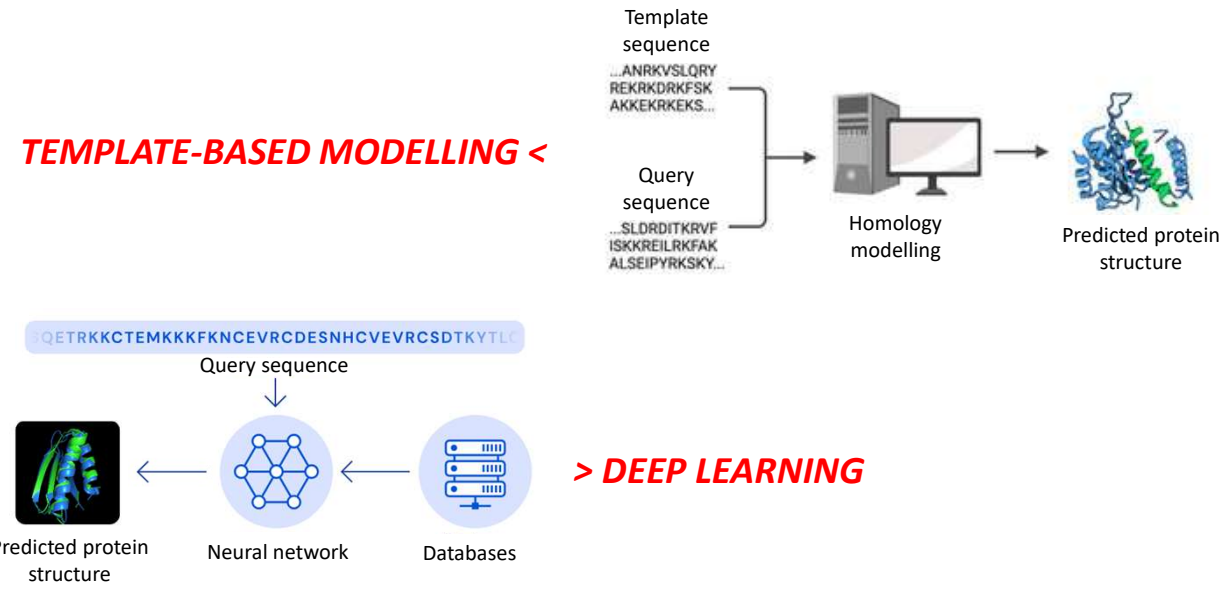
Structure Database



The atomic coordinates of most of these structures are deposited in a database known as the **Protein Data Bank (PDB)**: https://www.rcsb.org/. These data are readily available, which allows the structures of a variety of proteins to be analysed.

PDB-101 training materials (https://pdb101.rcsb.org/) help graduate students, postdoctoral scholars, and researchers use PDB data and RCSB PDB tools.

Structure Predictions

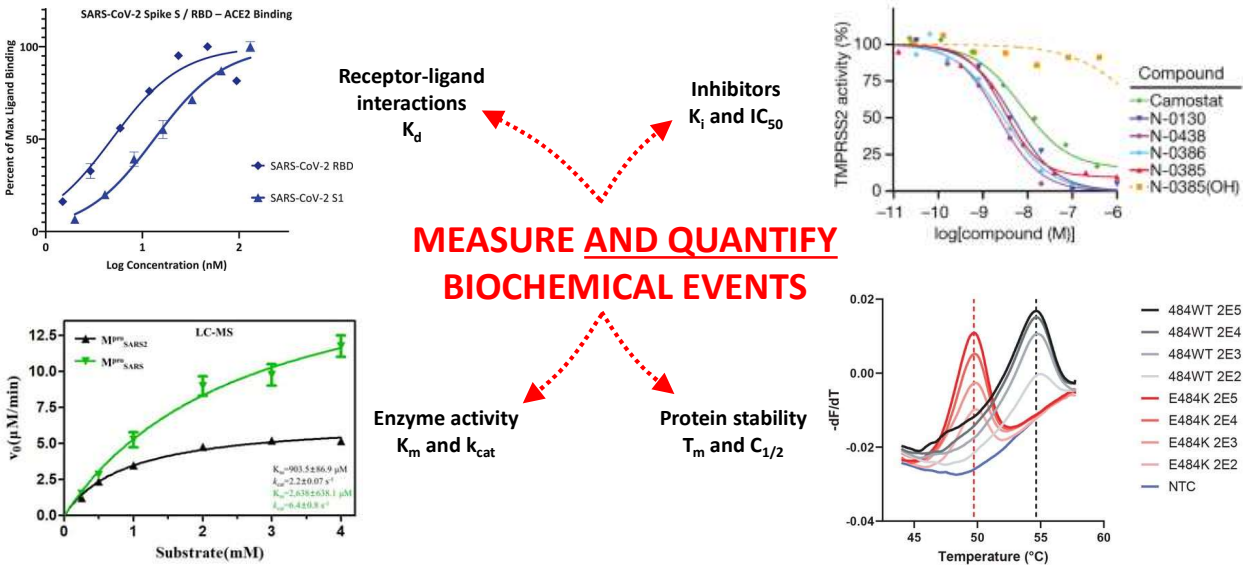


Figures: Advances in AI for Protein Structure Prediction: Implications for Cancer Drug Discovery and Development and AlphaFold: Using AI for scientific discovery - Google DeepMind

X-ray crystallographic, NMR and cryo-EM studies are often costly, inefficient, and time-consuming. At the same time, the number of known protein sequences far exceeds the number of experimentally determined structures, creating a gap that necessitates the use of computational approaches such as **homology modelling and deep learning**.

Homology modelling predicts a protein's 3D structure using the structure of a homologous protein. Deep learning uses a neural network to predict a protein's 3D structure.

Function

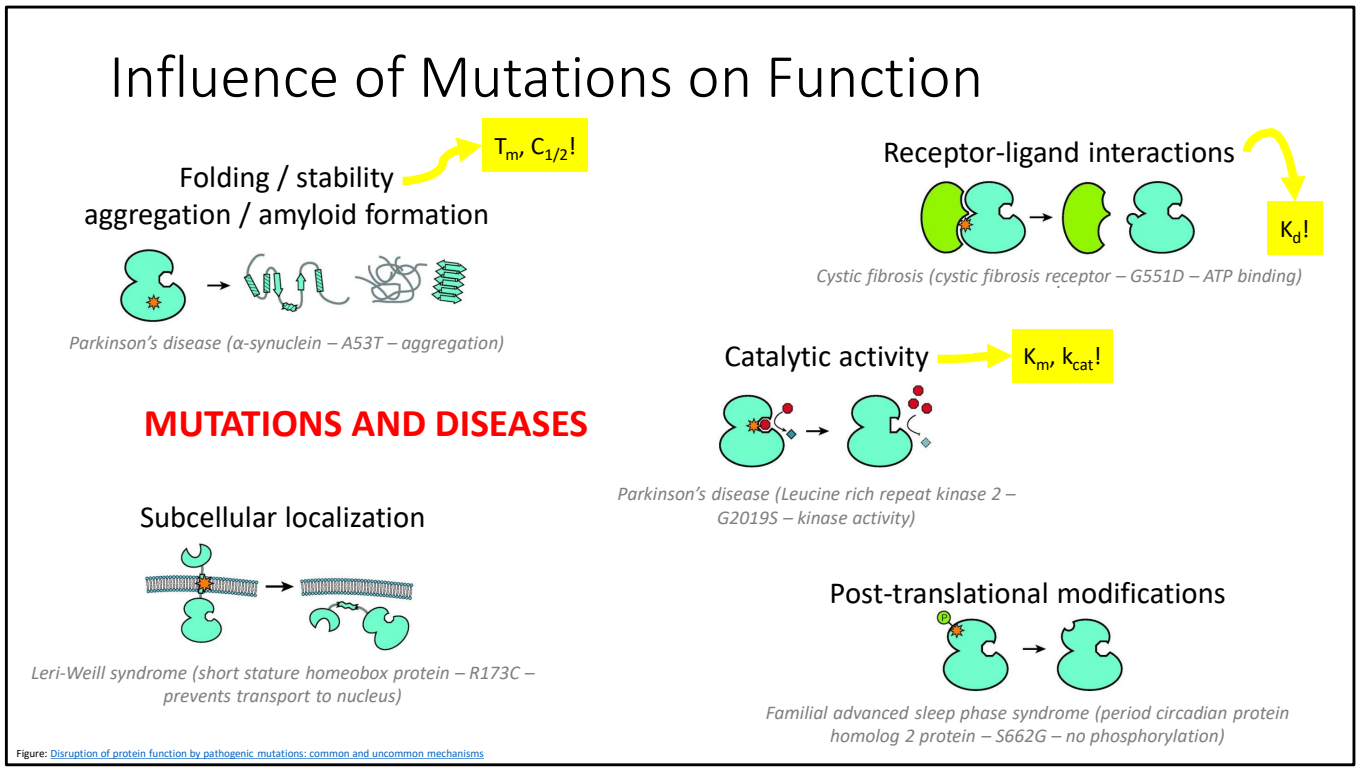


Figures: Molecules | Free Full-Text | Comprehensive Understanding of the Kinetic Behaviors of Main Protease from SARS-CoV-2 and SARS-CoV: New Data and Comparison to Published Parameters (mdpi.com), Frontiers | Methylene Blue Inhibits the SARS-CoV-2 Spike–ACE2 Protein-Protein Interaction–a Mechanism that can Contribute to its Antiviral Activity Against COVID-19 (frontiersin.org), A Novel Diagnostic Test to Screen SARS-CoV-2 Variants Containing E484K and N501Y Mutations | medRxiv and A TMPRSS2 inhibitor acts as a pan-SARS-CoV-2 prophylactic and therapeutic | Nature

Knowing a protein's structure is important, but understanding its **function** is essential for biology and medicine.

We use **quantitative methods** to measure how proteins interact, catalyze reactions, respond to inhibitors, and maintain stability:

- K_d (binding affinity), K_i (inhibition constant), and IC_{50} (half-maximal inhibitory concentration) are important for understanding protein-ligand interactions and designing therapeutics.
- K_m (Michaelis-Menten constant) and k_{cat} (turnover rate) help determine how efficiently an enzyme works and how it differs between variants
- T_m (melting temperature) and $C_{1/2}$ (concentration at half unfolding) are important for understanding mutation effects, vaccine design, and protein engineering.



There are several mechanisms by which mutations exert their effects:

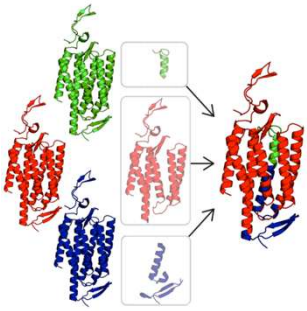
- **Folding, Stability, and Aggregation**
  - Example: Parkinson's disease / Mutation: A53T in α-synuclein / Effect: Promotes aggregation and amyloid formation
- **Receptor–Ligand Interactions**
  - Example: Cystic fibrosis / Mutation: G551D in cystic fibrosis receptor / Effect: Impairs ATP binding
- **Catalytic Activity**
  - Example: Parkinson's disease / Mutation: G2019S in LRRK2 kinase / Effect: Alters enzyme activity
- **Subcellular Localization**
  - Example: Leri-Weill syndrome / Mutation: R173C in SHOX protein / Effect: Prevents transport to the nucleus
- **Post-Translational Modifications**
  - Example: Familial advanced sleep phase syndrome / Mutation: S662G in PER2 protein / Effect: Prevents phosphorylation

How Can We Use This Knowledge?

DRUG DESIGN
To interfere with protein function



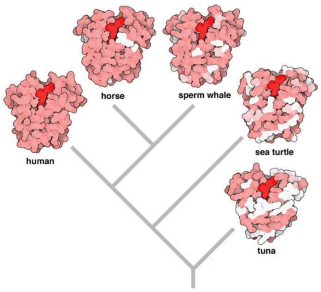
PROTEIN ENGINEERING
To modify substrate specificity, to improve protein stability ...



DE NOVO PROTEIN DESIGN
To design proteins with new structures and functions



EVOLUTION
To understand evolutionary relationships



Figures: photo by Anna Shvets: Yellow and White Red Pills on Blister Pack · Free Stock Photo (pexels.com), Methods in Protein Engineering - Arnold Group (caltech.edu), Data-driven protein design - Baker Lab, and PDB-101: Molecule of the Month: Globin Evolution

Our understanding of protein structure and function can be applied in real-world contexts across science, medicine, and technology:

- **Designing drugs** that interfere with protein function
- **Engineering proteins** to enhance or change their properties
- **Creating entirely new proteins** with novel structures and functions
- Using protein sequences and structures to trace **evolutionary relationships**

The Nobel Prize in Chemistry in 2018 went to protein engineering research, and the one in 2024 to protein structure prediction tools which pave the way for de novo design.

Evolution: Sequence Alignments and Motifs

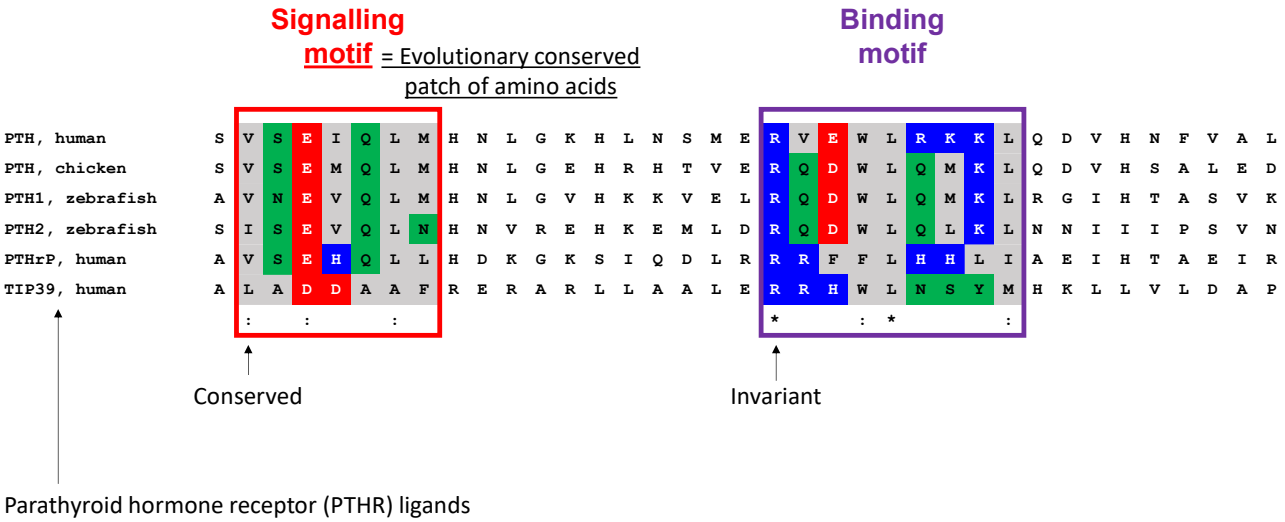


Figure: International Union of Basic and Clinical Pharmacology. XCIII. The Parathyroid Hormone Receptors—Family B G Protein–Coupled Receptors - ScienceDirect

Protein sequences change during **evolution** by **point mutations, insertions, and / or deletions** to the DNA. We can compare sequences, across species and within one species, via **multiple sequence alignments (MSAs)**:

- when the amino acids are identical between sequences, they are **invariant or absolutely conserved (shown by *)**
- when the amino acids are similar (*e.g.* negatively charged), they are **conserved (shown by :)**
- when there are many different residues, they are **not conserved**

Evolution conserves amino acids that are important to protein structure and function. Hence, knowledge of the primary structure of a protein can offer insights by searching for similarities with previously studied proteins using MSAs and looking for **evolutionarily conserved patches of amino acids to locate functional sites or structural elements, *i.e.* motifs**. Motifs are usually quite small, often only a few amino acids long. They include:

- active sites - amino acids involved in catalytic activity
- binding sites - amino acids that bind molecules
- post-translational modification sites - residues that are chemically modified after translation
- repeats - short amino acid sequences that are repeated within a protein and could be important for structural properties

Shown is a MSA of parathyroid hormone receptor (PTHR) ligands from human, zebrafish, and chicken. Amino acids in the signalling and binding motifs are coloured according to

their chemical properties: polar (G, S, T, Y, C, Q, N) - green; basic (K, R, H) - blue; acidic (D, E) - red; and hydrophobic (A, V, L, I, P, W, F, M) - grey. Invariant (*) and conserved (:) residues are highlighted.

Evolution: Consensus Sequence and Logo

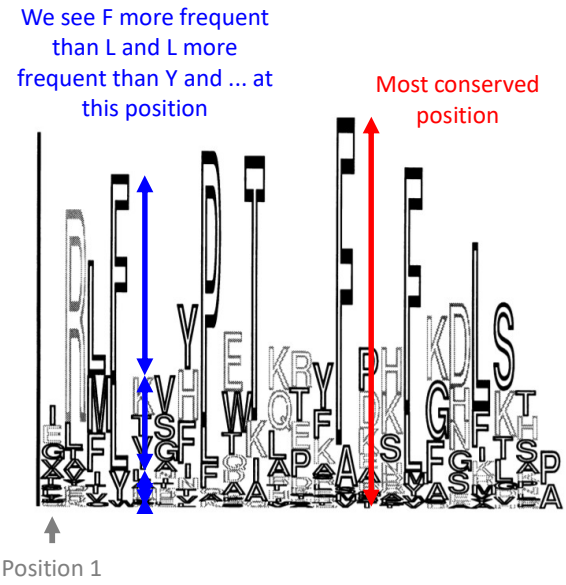


Figure: Sequence logos: a new way to display consensus sequences | Nucleic Acids Research | Oxford Academic

Typically, motifs are visualized with **consensus sequences**, often represented using **sequence logos**.

- Each logo consists of a stack of symbols for each position in the sequence.
- The overall height of the stack indicates the degree of conservation at that position.
- The height of each symbol in the stack indicates the relative frequency of that amino acid.
- The letters of each stack are ordered from most to least frequent, so that one may read the consensus sequence from the tops of the stacks.

Evolution: Phylogenetic Relationships and Homologs

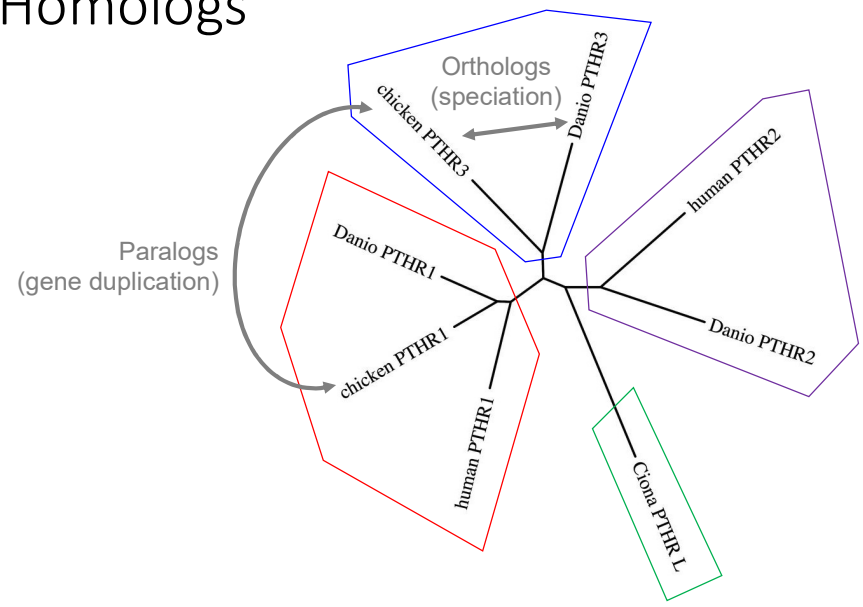


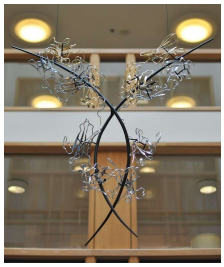
Figure: International Union of Basic and Clinical Pharmacology. XCIII. The Parathyroid Hormone Receptors—Family B G Protein–Coupled Receptors - ScienceDirect

We can reconstruct a **phylogenetic tree** using MSAs and our understanding of sequence evolution. Proteins that have diverged from a common ancestral gene are known as **homologs**:

- **Orthologs** are homologous proteins that are found in different species and evolved from a common ancestral gene by speciation. Often, orthologs retain the same function during evolution. *E.g.* PTHR1 in human and PTHR1 in chicken.
- Homologous proteins within the same genome are referred to as **paralogs**. Paralogs arise by gene duplication. They have often diverged in their function because the additional copy of the gene is redundant and usually free to evolve a new role or function. *E.g.* PTHR1 and PTHR2 in humans.

Shown are phylogenetic relationships among parathyroid hormone receptors (PTHRs) from different species. The diagram illustrates the three subtypes of PTHRs. PTH1R is present in all vertebrates, PTH2R is present in humans and zebrafish (*Danio rerio*) but not in birds, and PTH3R is present in birds and zebrafish but not in humans. The tree includes a PTHR-like sequence that was identified in the genome of the tunicate *Ciona intestinalis*. Branch distances indicate amino acid sequence divergence.

How Can We Use This Knowledge? *(cont.)*



Mike Tyka
Savior - IgG
2013



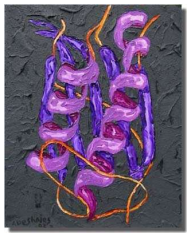
Irina Bezsonova
Capsaicin receptor (7LR0) – Spicy
2023



May K.
Adenylate kinase (1AEK) – Caesar
2012

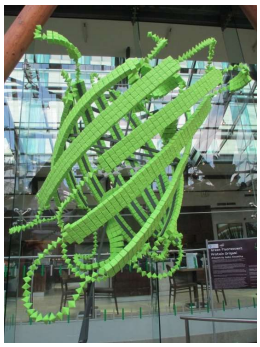


Julian Voss-Andreae
Unravelling collagen
2005



Jacques Deshaies
Alpha/Beta Proteome
2003

ART!



Seiko Kinoshita
GFP Origami
2015

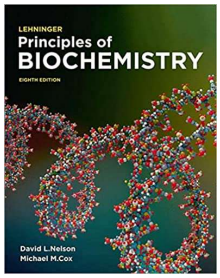
Figures: RCSB PDB Newsletter | Winter 2007, Painting proteins: Artworks by Jacques Deshaies (genomenewsnetwork.org), Protein Art | by May K. (wordpress.com), Seiko Kinoshita, textile artist • Work • GFP Origami (Green Fluorescent Protein), miketyka.com/?s=molecule and Inktober 2023 by Irina Bezsonova | Protein Data Bank in Europe

Proteins are not only functional molecules, they also have **beautiful, intricate structures**. Artists and scientists have collaborated to **visualize and interpret** these forms in sculptures, textiles, and paintings.

You Need to Be Able to...

- ❑ explain that there are four levels of protein structure and how they are distinguished from each other
- ❑ compare and identify the different types of covalent and noncovalent interactions involved in stabilising protein structure
- ❑ illustrate the importance of the hydrophobic effect
- ❑ look up information in the protein data bank (PDB)
- ❑ provide examples of protein structure-function relationships
- ❑ recall known effects of disease-associated mutations in proteins
- ❑ list the driving forces behind protein folding
- ❑ elaborate how cells can respond to unfolded, misfolded, and non-native proteins
- ❑ predict the effect of specific amino acid changes on protein structure and function
- ❑ describe and interpret multiple sequence alignments, sequence logos, and phylogenetic trees

Books – Essential and Recommended



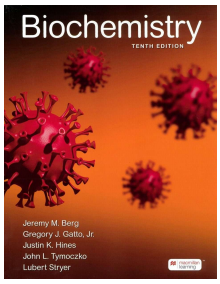
ESSENTIAL READING. Introductory core text.

Lehninger Principles of Biochemistry; David L. Nelson, Michael M. Cox

Publisher: Macmillan Learning, New York

Available online via UCL Library Services (through Kortext using your username@ucl.ac.uk and password)

Chapters 1, 2, 3, 4, 5, 6, and 12



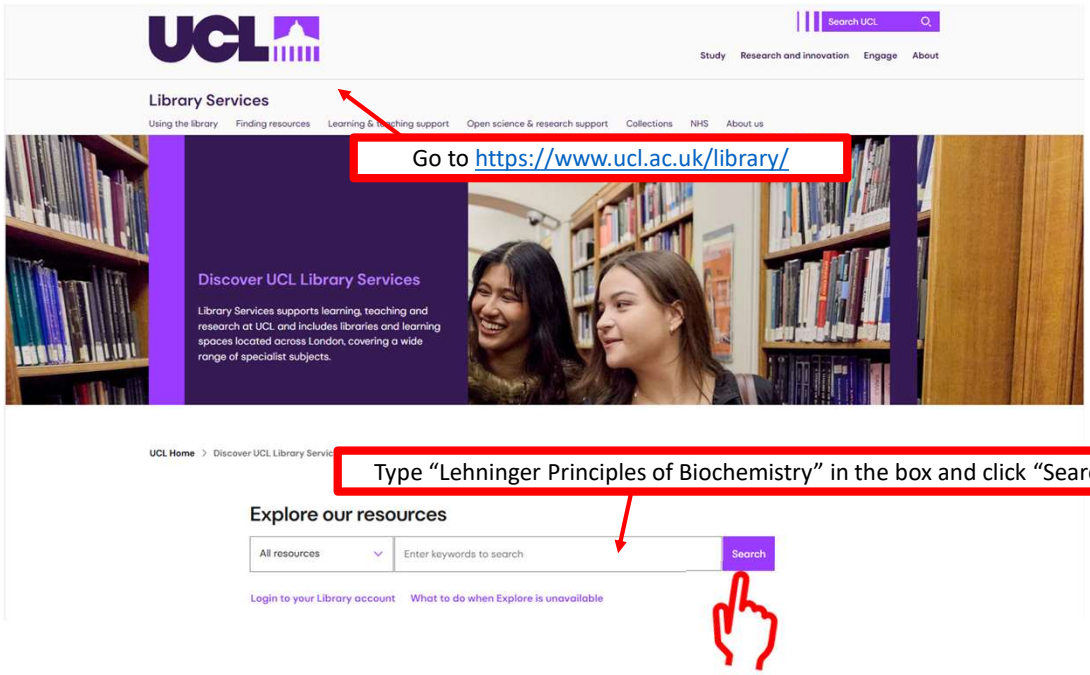
RECOMMENDED READING. Introductory text. Explained in a different way compared to Lehninger Principles of Biochemistry. Helpful to develop a full understanding.

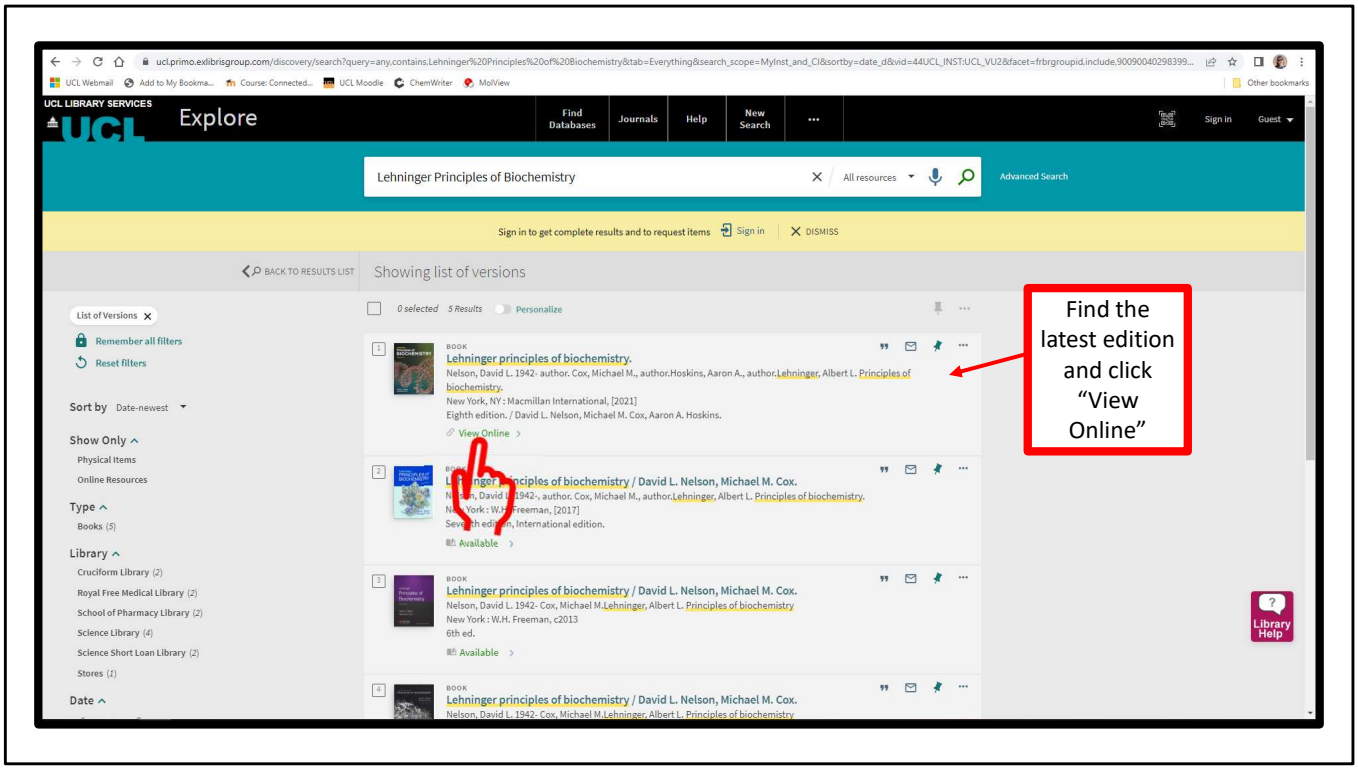
Biochemistry; Jeremy M. Berg, Gregory J. Gatto Jr., Justin K. Jines, John L. Tymoczko, Lubert Stryer

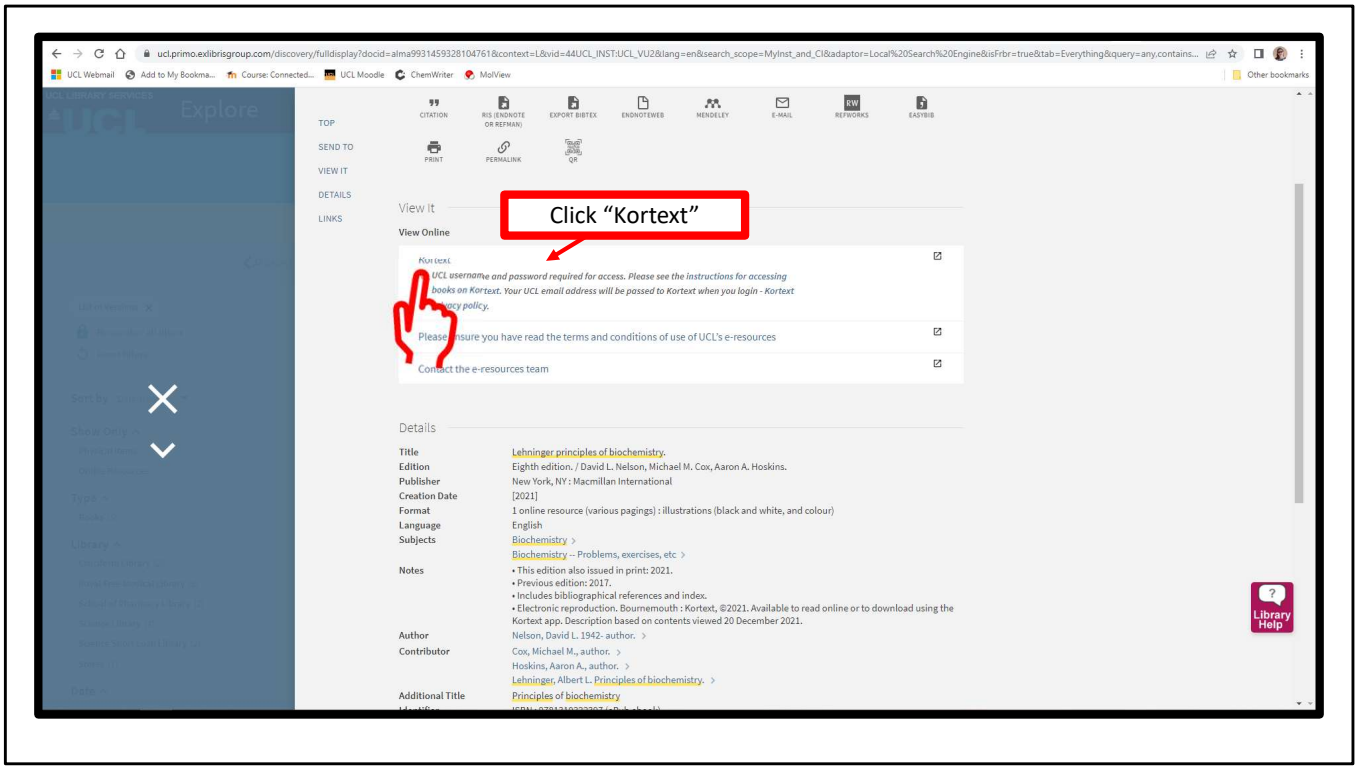
Publisher: Macmillan Learning, New York

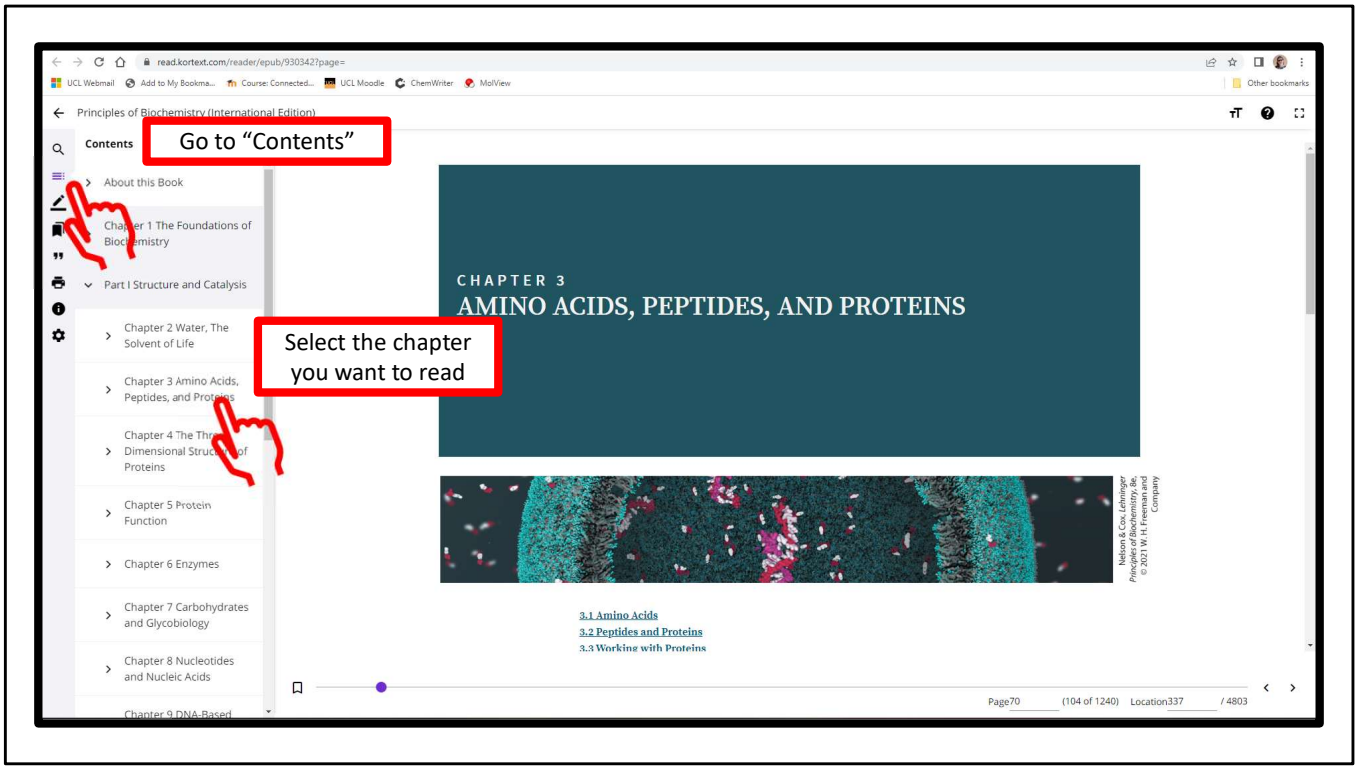
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Chapters 1, 2, 3, 5, 6, 7, 10, and 14

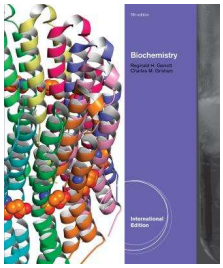

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Sign in
DISMISS
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Showing list of versions
List of Versions
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Show Only
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Online Resources
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Lehninger principles of biochemistry.
Nelson, David L. 1942- author. Cox, Michael M., author.Hoskins, Aaron A., author.Lehninger, Albert L. Principles of biochemistry.
New York, NY : Macmillan International, [2021]
Eighth edition. / David L. Nelson, Michael M. Cox, Aaron A. Hoskins.
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Nelson, David L. 1942- Cox, Michael M.Lehninger, Albert L. Principles of biochemistry
New York : W.H. Freeman, c2013
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Edition Eighth edition. / David L. Nelson, Michael M. Cox, Aaron A. Hoskins.
Publisher New York, NY : Macmillan International
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Format 1 online resource (various pagings) : illustrations (black and white, and colour)
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Biochemistry -- Problems, exercises, etc
Notes • This edition also issued in print: 2021.
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Author Nelson, David L. 1942- author.
Contributor Cox, Michael M., author.
Hoskins, Aaron A., author.
Lehninger, Albert L. Principles of biochemistry.
Additional Title Principles of biochemistry
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Chapter 1 The Foundations of Biochemistry
Part I Structure and Catalysis
Chapter 2 Water, The Solvent of Life
Chapter 3 Amino Acids, Peptides, and Proteins
Chapter 4 The Three-Dimensional Structure of Proteins
Chapter 5 Protein Function
Chapter 6 Enzymes
Chapter 7 Carbohydrates and Glycobiology
Chapter 8 Nucleotides and Nucleic Acids
Chapter 9 DNA-Based
Select the chapter you want to read
CHAPTER 3
AMINO ACIDS, PEPTIDES, AND PROTEINS
3.1 Amino Acids
3.2 Peptides and Proteins
3.3 Working with Proteins
Page 70 (104 of 1240) Location 337 / 4803

Books – Optional

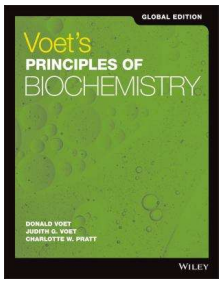


OPTIONAL READING. Introductory text. Explained in a different way compared to Lehninger Principles of Biochemistry. Helpful to develop a full understanding.

Biochemistry; Reginald H. Garrett, Charles M. Grisham

Publisher: Cengage Learning, Boston

In Science Library



OPTIONAL READING. Introductory text. Explained in a different way compared to Lehninger Principles of Biochemistry. Helpful to develop a full understanding.

Voet's Principles of Biochemistry; Donald Voet, Charlotte W. Pratt, Judith G. Voet

Publisher: John Wiley & Sons, Hoboken

In Science Library

Books – Optional *(cont.)*

Introduction to Protein Science → *Aimed at students interested in protein structure and function*

- Author: Arthur Lesk
- Third Edition ©2016
- Paperback ISBN: 9780198716846

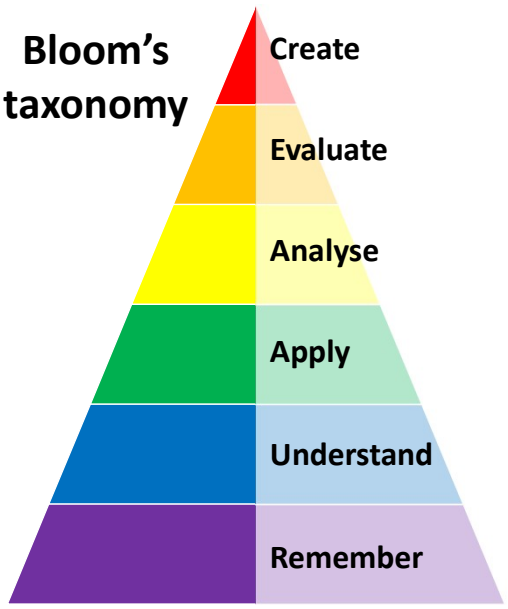
Proteins: Concepts in Biochemistry → *Aimed at students interested in biophysical chemical principles*

- Author: Paulo Almeida
- Second Edition ©2026
- Paperback ISBN: 9781032518404

Fundamentals of Enzyme Kinetics → *Aimed at students interested in enzyme kinetics*

- Author: Athel Cornish-Bowden
- Fourth Edition ©2012
- Paperback ISBN: 978-3-527-66549-5

How Do You Study?



Generate, modify, adapt, substitute, design ...

Assess, rank, criticize, justify, test, discriminate ...

Separate, arrange, classify, compare, debate ...

Calculate, complete, experiment, solve, relate, classify ...

Summarize, discuss, explain, draw, contrast ...

List, define, describe, match, name ...

See also: Microsoft Word - Exam Prep--How to Study With Bloom's Taxonomy.docx

Bloom's Taxonomy is a widely recognized hierarchical framework used by educators to classify and structure **educational objectives**. It categorizes learning into six progressive levels, from basic recall to complex creativity, helping students understand and achieve deeper learning.

For students, it provides a "toolbox" to track their own progress, starting with remembering facts, then understanding concepts, applying knowledge, analysing information, evaluating ideas, and finally creating new things. By working through these levels, students can develop a stronger, more comprehensive grasp of a subject.

To help you study the "Proteins & Enzymes" part of this module, we offer quizzes, a data analysis exercise, a workshop, and an online practical. Please see Moodle for more information.

Stay Up-to-date

- Nature News and Views →

https://www.nature.com/nature/articles?searchType=journalSearch&sort=PubDate&type=news-and-views

- Protein Data Bank "Molecule of the Month" →

https://pdb101.rcsb.org/motm/motm-by-date

- …

Try to keep up with developments in protein science beyond the lecture material. **Nature News and Views** is a good source of scientific perspectives, and the **Protein Data Bank's Molecule of the Month** pages are an accessible way to explore individual proteins and their biological functions.

These resources are opportunities to explore further rather than a list of additional material to memorise!

Communication

- Announcements
- Discussion for students
  - Keywords
- Student Office Hours (Fri 12 noon – 1 pm) → MS Booking Calendar link
- Personal (no lecture content!) → r.vancraenenbroeck@ucl.ac.uk